Response Tolerance and Spectral Regularity in Reference-Matched Reflectance Design: supplemental document

This document gives the continuous-function support bounds, mixture certificates, tolerance-interval proof, and response-cost calculation supporting the main article. It also proves the smooth-spectrum and input-continuity statements and derives the separate direction and norm comparisons. The accompanying exact fixture contains the rational spectra, response functions, and multipliers needed to verify the numerical inequalities without optimization.

1. INPUT FUNCTIONS AND EXACT REPRESENTATION

The response interval is $I = [380, 780]$ nm, with nodes $\lambda_i = 380 + i$ for $i = 0, \ldots, 400$. The CIE 1931 2° observer and the D65, A, and FL11 illuminants are specified by the datasets in Table S1. The FL11 column is selected from the fluorescent-illuminant dataset. Decimal tabulated values are read as exact rational numbers. Each observer and illuminant component is interpolated linearly between nodes.

Table S1. Source datasets. The selected interval and interpolation convention, followed by continuous white-Y normalization, define the mathematical model.

Input	Year	Dataset DOI
CIE 1931 observer	2019	10.25039/CIE.DS.xvudnb9b
D65	2019	10.25039/CIE.DS.hjfjmt59
A, 1 nm	2018	10.25039/CIE.DS.8jsxjrsn
Fluorescent illuminants, 1 nm	2018	10.25039/CIE.DS.54hy6srn

If E_j is an illuminant and $\bar{x}, \bar{y}, \bar{z}$ are the matching functions, then

$$f_j(\lambda) = \frac{100 E_j(\lambda)}{\int_I E_j \bar{y}} \begin{pmatrix} \bar{x}(\lambda) \\ \bar{y}(\lambda) \\ \bar{z}(\lambda) \end{pmatrix}. \tag{S1}$$

Every component of f_j is quadratic on each bin. The normalization is evaluated by exact integration of the product; it is not a rectangle sum over nodes. Positive scalar factors already present in a stored illuminant cancel in Eq. (S1).

Let N_i denote the standard continuous piecewise-linear hat functions. The exact nine-moment matrix is

$$T_{(j,k),i} = \int_I f_{j,k} N_i, \quad j = 0, 1, 2, \quad k = X, Y, Z. \tag{S2}$$

For a nodal spectrum $z = \sum_i z_i N_i$, its nine responses are Tz. The fixture contains a nonsingular nine-column minor and its rational inverse. Multiplication by the saved inverse is checked exactly. This verifies full row rank and supports the finite-dimensional corrections used below.

For each lower-bound witness, the verifier checks $|z_i| \le 1/2$ and the exact response equations. In class $\mathcal{S}$ it also checks $\ell \le z_{i+1} - z_i \le u$ on every 1 nm bin. These endpoint inequalities imply passivity and the derivative bounds on the complete piecewise-linear function. All numerical bounds concern the interpolated model of Eq. (S1).

2. MIXTURES AND NORMALIZATION

For $w_j \geq 0$, let $c_w = \sum_j w_j c_j$, $h_w = \sum_j w_j h_j$, and $e_w = \sum_j w_j e_j$. If K is attained by a named illumination and

$$K^2 c_i^{\mathsf{T}} c_j - h_i^{\mathsf{T}} h_j \geq 0 \quad \text{for every } i, j, \tag{S3}$$

then $\|h_w\|_2^2 \leq K^2 \|c_w\|_2^2$. Thus the same K is the maximum over the full cone. Similarly, the sufficient conditions

$$\rho^2 \beta^2 K^2 c_i^{\mathsf{T}} c_j - e_i^{\mathsf{T}} e_j \geq 0 \quad \text{for every } i, j \tag{S4}$$

prove the error condition for every mixture. These are sufficient copositivity certificates obtained through entrywise nonnegativity, not necessary conditions for arbitrary error vectors.

The scalar K^2, squared norms, and Gram entries are rational. All four X-direction witnesses used in the capacity comparison pass the appropriate mixture checks. The Z-direction comparison uses an exact lower witness, so its zero errors extend to mixtures directly. No approximate Z witness is needed for that comparison.

Named-lamp upper bounds are valid for the all-mixture problem because the latter includes the named conditions. Combined with mixture-feasible lower witnesses, this gives the same reported enclosures for the two domains. It does not identify their unknown exact capacities with one another.

3. SUPPORT CERTIFICATES ON CONTINUOUS FUNCTION CLASSES

A. Multiplier inequality and the unrestricted class

Write $h_j = Hc_j$ and $e_j = A_j z - \beta h_j$. For fixed multiplier vectors,

$$g = \sum_j \mu_j^{\mathsf{T}} f_j, \qquad P = \sum_j \mu_j^{\mathsf{T}} h_j. \tag{S5}$$

Then $\int_I gz = \beta P + \sum_j \mu_j^{\mathsf{T}} e_j$. The reference term is zero, while Cauchy–Schwarz bounds each remaining term below by $-\rho\beta K \|c_j\|_2 \|\mu_j\|_2$. For an outward upper bound W on their summed coefficients,

$$\beta(P - \rho W) \leq \int_I gz. \tag{S6}$$

Over $\mathcal{U}$, the support is exactly $\frac{1}{2}\int_I |g|$. Division by a positive $P - \rho W$ gives a rigorous upper capacity bound, irrespective of whether the multipliers are optimal.

For a polynomial $p(t) = \sum_{i=0}^n b_i B_i^n(t)$ in Bernstein form on $[0,1]$,

$$\frac{|\sum_i b_i|}{n+1} \leq \int_0^1 |p(t)|\, dt \leq \frac{\sum_i |b_i|}{n+1}. \tag{S7}$$

The right inequality uses nonnegativity of the basis; the left is the absolute value of the integral. Rational subdivision tightens these enclosures. A coordinate change to a physical bin of width h multiplies the integral by h. The verifier retains that factor on every subinterval, including the half-bin intervals of the cubic certificate.

B. Nodal support and the continuous correction

For nodal values $v = (v_0, \ldots, v_{400})$, let $(Dv)_i = v_{i+1} - v_i$. Let $a_i = \int_I gN_i$. For any multiplier vector η on the differences, put $q = a - D^{\mathsf{T}}\eta$. Then

$$a^{\mathsf{T}} v = \eta^{\mathsf{T}} Dv + q^{\mathsf{T}} v \leq \sum_i \max(\ell\eta_i, u\eta_i) + \frac{1}{2}\|q\|_1. \tag{S8}$$

This support inequality uses only the passive and slope restrictions. In particular, it imposes no response-equality condition on the interpolant to which it is applied.

For $z \in \mathcal{S}$, its endpoint interpolant Lz satisfies those restrictions. On one unit bin write $g(t) = a + bt + ct^2$ and $z(t) = z(0) + \int_0^t z'(s)ds$. Subtracting $(1-t)z(0) + tz(1)$ and exchanging integrals gives

$$\int_0^1 g(z - Lz) = \int_0^1 K_P(s) z'(s) ds, \tag{S9}$$

$$K_P(s) = \int_s^1 g(t)dt - \int_0^1 tg(t)dt \tag{S10}$$

$$= a/2 + b/6 + c/12 - as - bs^2/2 - cs^3/3. \tag{S11}$$

The kernel has zero integral. Subtract $(\ell + u)/2$ from z' inside the last integral to obtain

$$\left| \int_0^1 g(z - Lz) \right| \le \frac{u - \ell}{2} \int_0^1 |K_P|. \tag{S12}$$

Summing Eq. (S12) over all bins and adding Eq. (S8) bounds $\int_I gz$ for every absolutely continuous spectrum in the envelope. The exact-response capacity bound divides by $P > 0$. Numerically, the finite support contributes approximately 0.2101706137820 to the amplitude upper, and the continuous correction contributes approximately 0.00702177349656. Their certified sum is the upper B_0 used in the main theorem.

C. A continuous cubic support bound

Let ϕ be continuous and piecewise cubic, with $\phi(380) = \phi(780) = 0$. For every $z \in \mathcal{S}$,

$$\int_I gz = -\int_I \phi z' - \int_I (\phi' - g)z \tag{S13}$$

$$\le \frac{u - \ell}{2} \int_I |\phi| - \frac{u + \ell}{2} \int_I \phi + \frac{1}{2} \int_I |\phi' - g|. \tag{S14}$$

The first term follows by maximizing $-\phi z'$ pointwise over $z' \in [\ell, u]$; the second uses the passive box. The fixture supplies ϕ and μ. Exact continuity and endpoint checks validate the integration by parts. Polynomial differentiation, residual construction, and Bernstein integration then give the full-S tolerant upper bound. This upper need not equal the optimum of a nodal optimization problem.

4. ABSOLUTE BONUS AND TOLERANCE INTERVAL

A. Capacity arithmetic

Use the four capacity intervals in the main article and denote lower and upper endpoints by superscripts $-$ and $+$. Then, for example,

$$\Delta_{\mathcal{S}}(r) \ge b_{\mathcal{S}}^{-}(r) - b_{\mathcal{S}}^{+}(0)/(1 - r), \tag{S15}$$

$$\Delta_{\mathcal{U}}(r) \le b_{\mathcal{U}}^{+}(r) - b_{\mathcal{U}}^{-}(0)/(1 - r). \tag{S16}$$

At $r = 0.10$, exact cross-multiplication proves that the first lower bound exceeds 1.50 times the second upper bound. The reverse endpoint combinations give the remaining interval endpoints and establish a positive unrestricted bonus. The raw increments have enclosures

$$b_{\mathcal{S}}(r) - b_{\mathcal{S}}(0) \in (0.1855354858, 0.2762275169), \tag{S17}$$

$$b_{\mathcal{U}}(r) - b_{\mathcal{U}}(0) \in (0.1292643479, 0.1499960386). \tag{S18}$$

The lower difference is greater than 0.0355394472, proving a decrease of the raw regularity penalty. To obtain its relative decrease, use the following outward display bounds from Table 1 of the main article:

$$d(0) \ge 0.3817259867 - 0.2171923873 = 0.1645335994 =: d_0^{-}, \tag{S19}$$

$$0 \le d(0.10) \le 0.5317220254 - 0.4027278731 = 0.1289941523 =: d_1^{+}. \tag{S20}$$

Since $d_0^{-} > 0$ and $d_1^{+} \ge 0$,

$$\frac{d(0) - d(0.10)}{d(0)} = 1 - \frac{d(0.10)}{d(0)} \ge 1 - \frac{d_1^{+}}{d_0^{-}} = \frac{355394471}{1645335994} > 0.2160. \tag{S21}$$

This ratio calculation proves the claimed reduction of more than 21.60%. The full rational certificates give a slightly stronger lower bound, exceeding 0.21600115.

B. Feasible interpolation below the anchor tolerance

Let (z_0, b_0) be the exact S witness and (z_1, b_1) the witness at $r = 0.10$. For $0 \le t \le 1$, define

$$z_t = (1-t)z_0 + tz_1, \qquad \beta_t = (1-t)b_0 + tb_1. \tag{S22}$$

Convexity preserves the spectral constraints and reference equality. The error is t times that of z_1, so a sufficient tolerance is $x = trb_1/\beta_t$. Solving for t yields

$$\beta_t = \frac{b_0}{1-kx}, \qquad k = \frac{b_1 - b_0}{rb_1}, \qquad 0 \le x \le r. \tag{S23}$$

The same calculation is valid for the full mixture domain because both endpoint witnesses satisfy all of its constraints.

For $r \le x < 1$, keep z_1 and set $\beta = (1-r)b_1/(1-x)$. Then $\beta \ge b_1$, and

$$\|A_w z_1 - \beta h_w\|_2 \le [rb_1 + (\beta - b_1)]K\|c_w\|_2 = x\beta K\|c_w\|_2. \tag{S24}$$

This proves the upper-tolerance lower curve without a new solve.

Table S2. Constants used in the tolerance and norm comparisons. Values shown are decimal summaries; the rational fixture determines the signs in the proof.

Symbol	Meaning	Approximate value
b_0	exact S witness	0.2101706137814355
b_1	tolerant S witness	0.4027278731222646
B_0	exact S upper	0.21719238727858622
U_0	exact U lower	0.38172598678848113
L	unrestricted support upper	0.051541722172949694
P	target pairing for that support	0.13027912697583505
W	outward error coefficient	0.333455419161111
k	blend coefficient	4.781324367940393

C. Whole-interval sign proof

Set $D_0 = U_0 - B_0 > 0$. Below r, a lower bound for the bonus difference is

$$F(x) = \frac{b_0}{1-kx} - \frac{L}{P-Wx} + \frac{D_0}{1-x}. \tag{S25}$$

Multiplication by $(1-kx)(P-Wx)(1-x) > 0$ gives

$$q(x) = b_0(P-Wx)(1-x) - L(1-kx)(1-x) + D_0(1-kx)(P-Wx). \tag{S26}$$

Its coefficients are checked exactly: the constant is negative and the linear and quadratic coefficients are positive. Hence q is strictly increasing for $x \ge 0$. The rational sign $q(0.056736) > 0$ proves positivity throughout $[0.056736, r]$.

Above r, the lower difference is

$$\frac{(1-r)b_1 + D_0}{1-x} - \frac{L}{P-Wx}. \tag{S27}$$

Its numerator after multiplication by the positive denominators is affine and decreasing. It remains positive at $x = 0.137809$, proving the second half of the interval. The same rational checks verify denominator positivity throughout. The plot of these functions illustrates this argument but is not used to establish it.

5. UNIVERSAL ACTUAL RESPONSES AND THE SIGNED A-Y COST

A. Ball and half-space relaxation

For the exact S support let $L_{\mathcal{S}} = P_{\mathcal{S}} B_0$. At amplitude β, every feasible design satisfies

$$\mu_1^{\mathsf{T}} e_1 + \mu_2^{\mathsf{T}} e_2 \le L_{\mathcal{S}} - \beta P_{\mathcal{S}}. \tag{S28}$$

For $j \in \{1,2\}$ and the other index $o = 3 - j$, put

$$C_j = P_{\mathcal{S}} - \rho K \|c_o\|_2 \|\mu_o\|_2, \quad B_j = L_{\mathcal{S}} - \beta C_j, \quad T_j = \beta \rho K \|c_j\|_2. \tag{S29}$$

The error lies in $\{e : \|e\|_2 \le T_j,\ \mu_j^{\mathsf{T}} e \le B_j\}$. This set is a relaxation of the spectral problem. Any upper bound on a coordinate over it applies to every feasible spectrum.

For a unit vector a and nonzero μ, let $p = a^{\mathsf{T}}\mu$, $s^2 = \|\mu\|_2^2$. If $pT \le B$, the unconstrained ball maximizer Ta is feasible. Otherwise, at a nonempty cap with $-sT \le B < sT$, the optimum lies in the plane $\mu^{\mathsf{T}} e = B$. Decompose

$$e = \frac{B}{s^2}\mu + v, \qquad v \perp \mu, \qquad \|v\|_2 \le \sqrt{T^2 - B^2/s^2}. \tag{S30}$$

The projection of a into that plane has norm $\sqrt{1 - p^2/s^2}$, giving

$$\max a^{\mathsf{T}} e = \frac{pB + \sqrt{s^2 - p^2}\sqrt{s^2 T^2 - B^2}}{s^2}. \tag{S31}$$

An empty cap excludes feasibility altogether. Every branch choice used at the stated amplitude is checked using outward rational intervals.

B. Numerical consequences and extension to larger amplitudes

At $\beta_* = b_1$, the A values are approximately

$$\mu_1 = (0.00545967376, 0.04593605206, -0.00767597078)^{\mathsf{T}}, \tag{S32}$$

with $C_1 \simeq 0.09250594414$, $B_1 \simeq -0.01620148977$, and $T_1 \simeq 0.59630591471$. Evaluating the cap for $a = e_Y$ and $a = -e_X$ gives the necessary Y deficit and the actual-X lower bound in the main article. Under FL11, the X cap bound reduces to the ordinary error-ball bound, and no positive universal Y deficit follows.

Normalize by amplitude, $v_j = e_j/\beta$. The constraints become

$$\|v_j\|_2 \le \rho K \|c_j\|_2, \quad \mu_j^{\mathsf{T}} v_j \le L_{\mathcal{S}}/\beta - C_j. \tag{S33}$$

These sets decrease with β. At β_* the bound for the actual X response per unit amplitude is positive, and the A-Y upper per unit amplitude is negative. Multiplying by any larger feasible amplitude preserves and strengthens the absolute inequalities. This proves the all-feasible quantifier at or above the threshold. The exact-response comparator is the same class $\mathcal{S}$ for the same prescribed transformation: its X increments are bounded above by $B_0 h_{j,X}$.

If $e_{1,Y} \ge 0$ and $\mu_{1,Y} > 0$, the most negative support contribution in the A ball occurs with $e_{1,Y} = 0$. Therefore

$$\beta \le \frac{L_{\mathcal{S}}}{C_1 - \rho K \|c_1\|_2 \sqrt{\mu_{1,X}^2 + \mu_{1,Z}^2}} < 0.2679937276830. \tag{S34}$$

The denominator is certified positive. This bound establishes exclusion of the β_* regime when no A-Y loss is allowed, without claiming the bound is an attained fixed-Y optimum.

6. SMOOTH-SPECTRUM CAPACITY SUPREMA

Let Mz collect the nine response moments. The kernels are bounded on I and M has full rank on a finite piecewise-linear span. Choose nine functions with independent moment columns. Smooth compactly supported functions approximate them in $L^1(I)$, so boundedness of the kernels and determinant continuity provide fixed $\psi_i \in C_c^\infty(\operatorname{int} I)$ with invertible moment matrix V. Define

$$Rq = \sum_i \psi_i (V^{-1} q)_i. \tag{S35}$$

Then $MRq = q$, and there are finite constants C_0, C_1 such that $\|Rq\|_\infty \le C_0 \|q\|_\infty$ and $\|(Rq)'\|_\infty \le C_1 \|q\|_\infty$.

Proposition S1. *For every $z \in \mathcal{U}$ and $0 < \theta < 1$, there is a smooth $z_\infty \in \mathcal{U}$ with $Mz_\infty = \theta Mz$. If $z \in \mathcal{S}$, z_∞ can be chosen in $\mathcal{S}$.*

Proof. For $\mathcal{U}$, extend θz by zero and convolve with a nonnegative compactly supported smooth approximate identity. The result g_ε satisfies $|g_\varepsilon| \le \theta/2$ and converges in $L^1(I)$ to θz. Hence $q_\varepsilon = \theta Mz - Mg_\varepsilon \to 0$. The function $g_\varepsilon + Rq_\varepsilon$ has exactly the desired moments and stays inside the passive box for sufficiently small ε.

For $\mathcal{S}$, extend z constantly beyond the endpoints. Its derivative remains in $[\ell, u]$, since $0 \in [\ell, u]$. Mollification of θz yields $\theta\ell \le g_\varepsilon' \le \theta u$, preserves the contracted passive box, and converges uniformly. Choose ε small enough that

$$C_0\|q_\varepsilon\|_\infty < (1-\theta)/2, \qquad C_1\|q_\varepsilon\|_\infty < (1-\theta)\min(-\ell, u). \tag{S36}$$

The correction then fits both spectral margins and restores all nine moments exactly. □

The response constraints are positively homogeneous in (Mz, β). Thus $(z_\infty, \theta\beta)$ is feasible whenever (z, β) is feasible. All mixture responses follow from the exact generator moments. Taking $\theta \uparrow 1$, and using inclusion for the reverse inequality, proves equality of the original and smooth-subclass capacity suprema. The construction does not impose a common second-derivative bound or prove smooth attainment at an endpoint amplitude.

7. CONTINUITY WITH RESPECT TO INPUT FUNCTIONS

For each input parameter p, define the stacked moment operator, target, and generator-error vector by

$$M_p z = (A_{p,0}z, A_{p,1}z, A_{p,2}z)^{\mathsf{T}} \in \mathbb{R}^9, \tag{S37}$$

$$h_p = (h_{p,0}, h_{p,1}, h_{p,2})^{\mathsf{T}}, \qquad e = M_{p_0}z - \beta h_{p_0}. \tag{S38}$$

Each displayed component block has three coordinates. For mixture weights w, write $e(w) = \sum_j w_j e_j$, and define $c_p(w)$ and $h_p(w)$ by the same generator sums.

Proposition S2 (Continuity at a fixed tolerance)**.** *Fix $0 \le \rho < 1$ and one of the two illumination domains. Suppose the response kernels converge componentwise in $L^1(I)$ as $p \to p_0$, the targets vary continuously with $h_{p,0} = 0$, and the target at p_0 is nonzero. Assume that M_{p_0} has rank nine and that incident-response norms are uniformly positive on the selected compact illumination domain in a neighborhood of p_0. Keep the spectral classes fixed, with $\ell < 0 < u$. Then, for $\mathcal{C} = \mathcal{U}, \mathcal{S}$, $b_{\mathcal{C},p}(\rho) \to b_{\mathcal{C},p_0}(\rho)$. The bonus difference is therefore continuous at every fixed $\rho < 1$.*

Proof. Uniform convergence of $h_p(w)$ and $c_p(w)$ on the illumination domain makes K_p continuous and locally positive.

The preceding smooth basis has moment matrices $V_p \to V_{p_0}$. Thus $R_p q = \sum_i \psi_i (V_p^{-1} q)_i$ is a locally uniformly bounded right inverse in both the function and derivative norms. Small positive multiples of $R_p h_p$ are exact feasible designs, ensuring positive exact capacity nearby.

For upper semicontinuity, the triangle inequality at an illumination attaining K_p provides a uniform upper bound on feasible amplitudes. In $\mathcal{U}$, weak-star compactness gives a subsequential limit; L^1 kernel convergence and bounded spectra imply convergence of moments. In $\mathcal{S}$, boundedness and a common Lipschitz constant give uniform subsequential convergence. The inequalities $\ell(y-x) \le z(y) - z(x) \le u(y-x)$ pass to that limit. In either class the reference and every fixed-mixture response constraint pass to the limit, proving

$$\limsup_{p \to p_0} b_{\mathcal{C},p}(\rho) \le b_{\mathcal{C},p_0}(\rho). \tag{S39}$$

At $\rho = 0$, contract an exact feasible design by $\theta < 1$ and correct its moments with

$$q_p = \theta\beta h_p - M_p(\theta z), \qquad z_p = \theta z + R_p q_p. \tag{S40}$$

The correction tends to zero and fits the strict spectral margins. This yields lower semicontinuity at exact tolerance.

At $0 < \rho < 1$, first blend a feasible pair (z, β) with a positive exact pair (z_e, β_e) in the same class:

$$\bar{z} = (1-\delta)z + \delta z_e, \qquad \bar{\beta} = (1-\delta)\beta + \delta\beta_e. \tag{S41}$$

The error becomes $(1-\delta)e(w)$, and its cone slack is at least $\rho\delta\beta_e K_{p_0}\|c_{p_0}(w)\|_2$, uniformly positive on the simplex. Contract by $\theta < 1$ for spectral slack, then prescribe the generator moments

$$d_p = \theta[\bar{\beta}h_p + (1-\delta)e], \quad q_p = d_p - M_p(\theta\bar{z}), \quad z_p = \theta\bar{z} + R_p q_p. \tag{S42}$$

The corrected spectrum has reference error zero and mixture errors $\theta(1-\delta)e(w)$. Continuity preserves the strict cone slack, and the small correction fits the spectral margins. Letting $\theta \uparrow 1$ and $\delta \downarrow 0$ proves lower semicontinuity. Capacities, and hence the bonus difference at fixed ρ, are continuous. □

The strict central inequality persists in some neighborhood of the frozen inputs. This argument gives no numerical radius for measurement or interpolation uncertainty.

8. SEPARATE DIRECTION AND NORM COMPARISONS

A. The X and Z directions in common units

For $d = X, Z$, define $\Phi_d = K_d[b_{\mathcal{S},d}(0.10) - b_{\mathcal{S},d}(0)/0.9]$. Target inflation gives $\Phi_d \geq 0$. The X lower is the central S bonus lower multiplied by an outward lower enclosure of K_X. The Z upper is the full-AC tolerant upper, obtained by the nodal support and Peano correction, minus an exact Z witness amplitude divided by 0.9, then multiplied by an outward upper enclosure of K_Z. Let these two positive certificate endpoints be A and B. They satisfy

$$A > 0.0156453680148, \qquad B < 0.0003455399445, \qquad A > 45.278B > 0. \tag{S43}$$

Consequently $\Phi_X > 45.278\Phi_Z$. This statement remains valid when the unknown actual Z bonus is zero; no ratio with a zero actual bonus is taken.

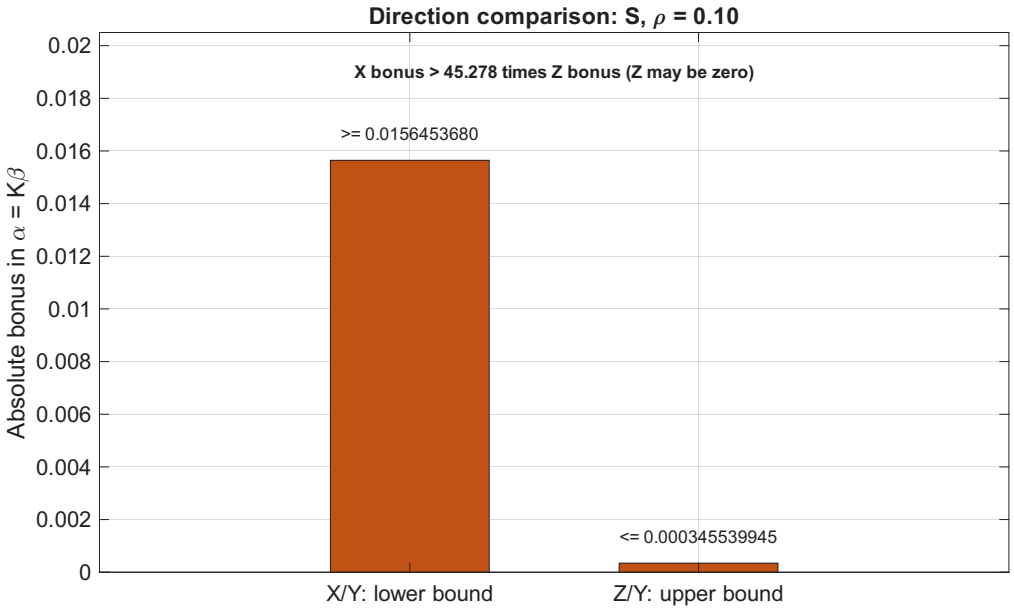


Fig. S1. Euclidean direction comparison in common $\alpha = K\beta$ units. The X lower bonus exceeds 45.278 times the Z upper bonus. This comparison concerns two fixed transformations and is separate from the norm perturbation result.

B. A sufficient linear-norm radius

Let D be an invertible real 3×3 matrix used on error, target, and incident response. With $\kappa = \mathrm{cond}_2(D)$, singular-value bounds imply $K_I/\kappa \leq K_D \leq \kappa K_I$. The normalized error criterion J_D therefore satisfies

$$\kappa^{-2} J_I \leq J_D \leq \kappa^2 J_I. \tag{S44}$$

For $1 \leq \kappa^2 \leq t$ and $rt < 1$, this yields the finite-capacity comparison

$$b_{\mathcal{C},I}(r/t) \leq b_{\mathcal{C},D}(r) \leq b_{\mathcal{C},I}(rt), \qquad b_{\mathcal{C},D}(0) = b_{\mathcal{C},I}(0). \tag{S45}$$

The exact baseline is unchanged by an invertible coordinate map. The inflation subtraction remains $b_{\mathcal{C}}(0)/(1-r)$ in both comparisons. Using Table S2,

$$\Delta_{\mathcal{S},D}(r) \geq b_0/(1-kr/t) - B_0/(1-r), \tag{S46}$$

$$\Delta_{\mathcal{U},D}(r) \leq L/(P-rWt) - U_0/(1-r). \tag{S47}$$

Multiplying the difference by $(1-r)(t-kr)(P-rWt) > 0$ gives

$$q_D(t) = b_0 t(P-rWt)(1-r) - L(t-kr)(1-r) + D_0(t-kr)(P-rWt). \tag{S48}$$

The quadratic coefficient is negative, $q_D'(1) < 0$, and $q_D(1) > 0$. Exact rational evaluation also gives $q_D(1.052918^2) > 0$, with all denominators positive over that interval. Since the derivative decreases, the polynomial is positive throughout $1 \leq t \leq 1.052918^2$. This proves the sufficient condition-number bound in the main article. At $\kappa \leq 1.02$, the lower S and upper U bounds are approximately 0.1475673909 and 0.1150759345. No assumption is made that the lamp attaining K_D remains unchanged.

9. REPRODUCTION AND VERIFICATION STATUS

The accompanying directory `reproducibility/` is a self-contained subset of the research files. It contains the exact fixture, the verified display export, a standard-library Python checker, and a MATLAB presentation function. From that directory, run

```
python -B tools/verify_paper_results.py
```

The default is read-only. It reconstructs the integrated model, checks rank, verifies the five selected X/Z witnesses, evaluates full-class supports, and checks the absolute, interval, cost, and scope inequalities. It needs no optimizer, network access, machine configuration, or historical project imports. Assertions must remain enabled; the script rejects Python optimization mode.

For an explicitly writable export followed by presentation checks, use

```
python -B tools/verify_paper_results.py --emit results/research/R135-paper-verified.json
```

and, from MATLAB in the same directory,

```
run_paper_results('verify')
run_paper_results('render')
```

Rendering uses the verified scalar export and checks its fixture hash. It creates deterministic plots and tables without optimization. The saved rendering was produced with MATLAB R2025b; the exact checker was verified with Python 3.12.14. Plot coordinates and response summaries use floating-point conversion. Capacity and bonus CSV endpoints are generated as outward decimal strings directly from exact rational bounds. The certificates use rational arithmetic and outward radical enclosures; the verifier also checks the displayed 21.60% calculation and the printed response inequalities.

The original tolerant S solver's non-success flag is retained in the fixture. Its lower bound is justified by the subsequently repaired exact feasible spectrum, not by a claim of successful optimizer termination. The support endpoints are rigorous bounds, not necessarily attained extrema. The smooth-spectrum and continuity propositions are analytical proofs; the arithmetic checker is not a proof-assistant formalization of them.

The frozen fixture's SHA-256 digest is

```
89c4a74ed3c54a05df04d8176cc3e8105
dcb84040799718314ab5246bcfd58d7.
```

The line break and final punctuation are not part of the digest. The package preserves input and source hashes, source dataset attribution, and the history of the numerical certificates. Author details and publication declarations are maintained separately from these technical files.